Q1. What is a Class in Python, and how is it used to create Objects?

Ans:

A class in Python is a blueprint or template used to create objects. It defines the properties (attributes) and behaviours (methods) that its objects will have.

An object is an instance of a class. We create objects by calling the class name.

Syntax:

```
class ClassName:
    # attributes and methods


object_name = ClassName()
```

Example:

```
class Student:
    def __init__(self, name, age):
        self.name = name
        self.age = age


student1 = Student("Apurva", 22)


print(student1.name)
print(student1.age)
```

Output:

```
Apurva
22
```

Explanation:

- class Student – Defines a class named Student.
- __init__() – A special method that initializes object attributes.
- self – Refers to the current object.
- name and age – Attributes of the class.
- student1 – An object created from the Student class.

Real-life example: A class is like a house blueprint, and objects are the actual houses built using that blueprint.

Q2. What are Methods and Attributes in Python Classes?

Ans:

In Python classes, attributes and methods are used to define the properties and behaviours of objects.

1. Attributes:

Attributes are variables defined inside a class or its objects. They store information or data about an object.

Example:

```
class Student:
    def __init__(self, name, age):
        self.name = name
        self.age = age


student1 = Student("Apurva", 22)


print(student1.name)
print(student1.age)
```

Output:

Apurva

22

Here, name and age are attributes.

2. Methods:

Methods are functions defined inside a class. They describe the actions or behaviours that an object can perform.

Example:

```
class Student:
    def study(self):
        print("Student is studying")


student1 = Student()
student1.study()
```

Output:

Student is studying

Here, study() is a method.

Difference between Attributes and Methods

Attributes	**Methods**
Attributes are variables.	Methods are functions.
They store data or information.	They perform actions.
They describe properties of an object.	They describe behaviours of an object.
Example: name, age, marks.	Example: study(), display(), calculate().

Remember:

- Class = Blueprint.
- Object = Instance of a class.
- Attributes = What an object has.
- Methods = What an object does.